Anoop Singh Sachdev

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Education

Thapar Institute of Engineering and Technology (TIET), Patiala, India Expected May 2027
Bachelor of Engineering, Computer Engineering CGPA: 7.6
Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Data Engineering, Database Management Systems, Software Engineering, Machine Learning, Artificial Intelligence, Cloud & DevOps

Delhi Private School, Sharjah, UAE 2022 – 2023
Class XII (SSCE) 93.6%

Delhi Private School, Sharjah, UAE 2020 – 2021
Class X (SSE) 94.8%

Technical Skills

Languages: Python, C++, JavaScript, SQL
Full-Stack & Backend: React.js, Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
Cloud & DevOps: Google Cloud Platform (Cloud Run, Compute Engine, BigQuery, Cloud Storage), Docker, Linux
Databases & Data Systems: MySQL, MongoDB, Redis, Apache Kafka
AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, ONNX, SHAP, Pandas, NumPy
AI-Assisted Development: Claude Code, Cursor (AI-assisted design, code generation, testing, and debugging)
Testing & Tools: Git, GitHub, Postman, Selenium, TestNG

Projects

ShopSense Python, FastAPI, Gemini (Vertex AI), Docker, GCP

- Built hybrid product search (BM25 + semantic embeddings, rank fusion, cross-encoder re-ranking) over 48,862 products from Amazon's ESCI dataset, lifting NDCG@10 from 0.468 to 0.574 on 500 held-out queries with human relevance labels.
- Added Gemini query understanding and LLM re-ranking on Vertex AI, reaching 0.600 NDCG@10 versus 0.565 for the cross-encoder, but at about 50 s versus 0.6 s per query and $1.14 per 1,000 queries, so the cross-encoder became the live default.
- Deployed a FastAPI service and web UI to Google Cloud Run with rate limiting and 189 pytest tests.

BeanSight PyTorch, YOLOv8, ONNX, FastAPI, Docker, GCP

- Built and deployed a full-stack web application that classifies green coffee beans into 17 defect categories (EfficientNet-B0) and detects defects in roasted beans (YOLOv8), turning manual quality inspection into a single image upload.
- Exported both PyTorch models to ONNX for lightweight CPU inference and served them through a Dockerized FastAPI REST backend.
- Automated build and deployment to Google Cloud Run with a GitHub Actions CI/CD pipeline.

Project Camp React, Node.js, Express.js, MongoDB

- Built a full-stack collaborative Kanban application with REST APIs for projects, task tracking, and role-based access control, using Mongoose schemas tuned for CRUD performance.
- Implemented secure authentication with JWT access and refresh tokens, email verification, and password reset via Nodemailer; integrated Cloudinary for file storage and deployed on Vercel.

Answer Gap Python, SQL, Statsmodels, scikit-learn

- Framed a marketplace product problem (which new questions go unanswered for 24 hours) and built the label and post-time features in SQL over 1.26M Stack Overflow questions, avoiding data leakage.
- Compared logistic regression, random forest, gradient boosting, and a title-word model on a time-based split; the title-word model ranked a 5% review queue at 69.5% precision versus a 40.9% base rate, with calibration drift monitored as the answer rate fell from 92% (2008) to 59% (2016).
- Ran pre-specified hypothesis tests with 95% confidence intervals (code blocks raise the answer rate by 3.5 points) and designed a follow-up A/B test: user-level randomization, about 18,000 questions per arm over 14 days, and a posting-volume guardrail metric.

Student Dropout Early-Warning System Python, Pandas, scikit-learn, LightGBM, SHAP

- Built an early-warning pipeline over 32,593 registrations and 10.6M learning-platform records with leakage-safe features; LightGBM on a time-based split reached 0.722 PR-AUC and 35.5% recall within a 20% outreach budget.

Certifications

- **Neural Networks and Deep Learning** – DeepLearning.AI, Coursera: backpropagation and gradient descent from scratch in NumPy.
- **Convolutional Neural Networks (In Progress)** – DeepLearning.AI, Coursera: image classifiers in TensorFlow/Keras with transfer learning (MobileNetV2).
- **Complete Web Development Course** – Udemy: React.js, Node.js, Express.js, MongoDB, REST APIs.